

March 9, 2026

Editors

*Mathematical Programming Computation*

Dear Editors,

We are pleased to submit our manuscript entitled “Practical Regularized Quasi-Newton Methods with Inexact Function Values” for publication in *Mathematical Programming Computation*.

This paper studies smooth nonconvex optimization problems in which function evaluations are affected by numerical errors. The main contributions of the paper are as follows:

- We develop a regularized quasi-Newton framework for smooth nonconvex optimization with noisy objective function evaluations.
- By combining a relaxed Armijo-type line search with a regularization parameter update inspired by Objective-Function-Free Optimization and AdaGrad-Norm, we establish a global convergence rate of  $\mathcal{O}(1/\varepsilon^2)$  for obtaining a first-order stationary point.
- Numerical experiments on the CUTEst benchmark set with injected noise and low-precision floating-point arithmetic (64-, 32-, and 16-bit) demonstrate that the proposed method is substantially more robust and faster than several existing methods.

We believe that this work will be of interest to researchers in numerical optimization, particularly those studying quasi-Newton methods and optimization algorithms under noise.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to this journal.

Thank you for considering this manuscript.

Sincerely,

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